

Noisy Truth:WIDS Project Report

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1 Introduction

State estimation is a central problem in mobile robotics, where the objective is to infer the pose of a robot from noisy sensor measurements and a mathematical model of the system. Due to nonlinear robot dynamics and uncertainty in sensing and actuation, probabilistic estimation techniques are required.

In this project, an Unscented Kalman Filter (UKF) is implemented for localization of a differential drive robot in a simulated environment using ROS 2. The UKF is chosen because it handles nonlinear motion models without requiring linearization, unlike the Extended Kalman Filter.

The project consists of three main components:

- A differential drive simulator generating ground truth and noisy measurements
- A custom UKF implementation
- A ROS 2 node integrating simulation data with the filter

2 Noisy Data Simulation

2.1 Differential Drive Simulator

A custom ROS 2 node simulates the motion of a differential drive robot following a circular trajectory under constant linear and angular velocity

commands. The simulator publishes:

- Ground truth odometry
- Noisy odometry measurements

2.2 Noise Modeling

Noise is added to the ground truth pose to generate realistic sensor measurements. The noise is modeled as zero-mean Gaussian noise added independently to position and orientation.

Gaussian noise is chosen because in real systems, uncertainty arises from many independent factors such as wheel slip, encoder quantization, surface irregularities, and timing jitter. By the Central Limit Theorem, the combined effect of many independent random disturbances is well-approximated by a Gaussian distribution.

Mathematically:

$$\mathbf{z} = \mathbf{x}_{true} + \mathbf{v}, \quad \mathbf{v} \sim \mathcal{N}(0, R)$$

The simulator generates noisy data, while the UKF models how uncertain that data is using the measurement noise covariance matrix.

3 Unscented Kalman Filter Implementation

3.1 State and Models

The system state is defined as:

$$\mathbf{x} = [x, y, \theta]^T$$

The motion model is based on standard differential drive kinematics and is nonlinear in the state variables. Measurements directly observe all state variables with noise.

3.2 Custom UKF Class

A generic UKF class was implemented to allow flexibility and clarity. The class includes:

- **State vector and covariance:** mean state estimate and associated uncertainty
- **Sigma point generation:** captures mean and covariance using deterministically chosen points
- **Prediction step:** propagates sigma points through the nonlinear motion model
- **Update step:** incorporates measurements using probabilistic weighting

Key parameters of the UKF include:

- Initial covariance P_0 : encodes initial uncertainty
- Process noise covariance Q : models uncertainty in the motion model
- Measurement noise covariance R : models sensor uncertainty

These parameters control how the filter balances trust between prediction and measurement.

4 UKF Node Integration

The UKF is wrapped inside a ROS 2 node that subscribes to noisy odometry measurements and publishes estimated pose information. Each callback performs:

1. Prediction using the known control inputs
2. Measurement update using noisy odometry
3. Publication of the estimated pose

The prediction timestep is matched with the simulator update rate to ensure consistent state propagation.

5 Predicted Behavior Before Simulation

Before running simulations, the expected behavior of the filter under different parameter configurations can be predicted based on estimation theory.

5.1 Case A: Nominal Parameters

With balanced values of P_0 , Q , and R , the filter is expected to:

- Show initial deviation due to incorrect initialization
- Rapidly converge to the true trajectory
- Produce smooth and stable estimates

This configuration represents a reasonable balance between model trust and sensor trust.

5.2 Case B: Large Initial Covariance

Increasing P_0 increases the initial spread of sigma points. This is expected to:

- Cause larger initial uncertainty
- Allow faster correction using measurements
- Produce identical steady-state behavior

This configuration is realistic when the initial pose estimate is unreliable.

5.3 Case C: Small Measurement Noise

Reducing R causes the filter to trust measurements more strongly. This is expected to:

- Reduce smoothing
- Introduce visible jitter in the estimate
- Follow noisy measurements closely

Such a configuration is appropriate only when sensors are highly accurate and reliable.

5.4 Case D: Large Process Noise

Increasing Q models greater uncertainty in the motion model. This is expected to:

- Reduce overconfidence in prediction
- Smooth out aggressive corrections
- Rely more on measurements over time

This configuration is meaningful when the robot experiences unmodeled disturbances such as slip or uneven terrain.

6 Simulation Results

6.1 Case A: Nominal Configuration

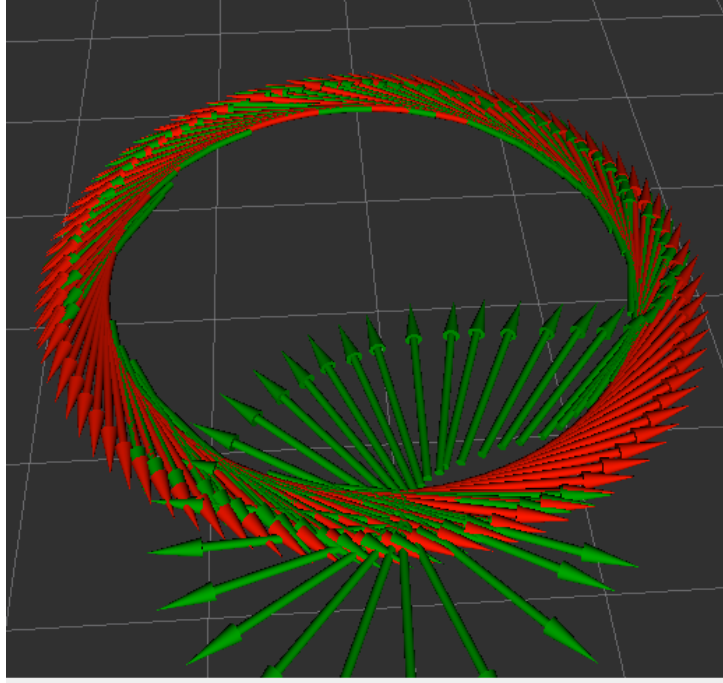


Figure 1: Case A: Balanced parameters resulting in smooth and stable convergence

The estimated trajectory converges rapidly to the ground truth and remains stable throughout the motion.

6.2 Case B: Increased Initial Covariance

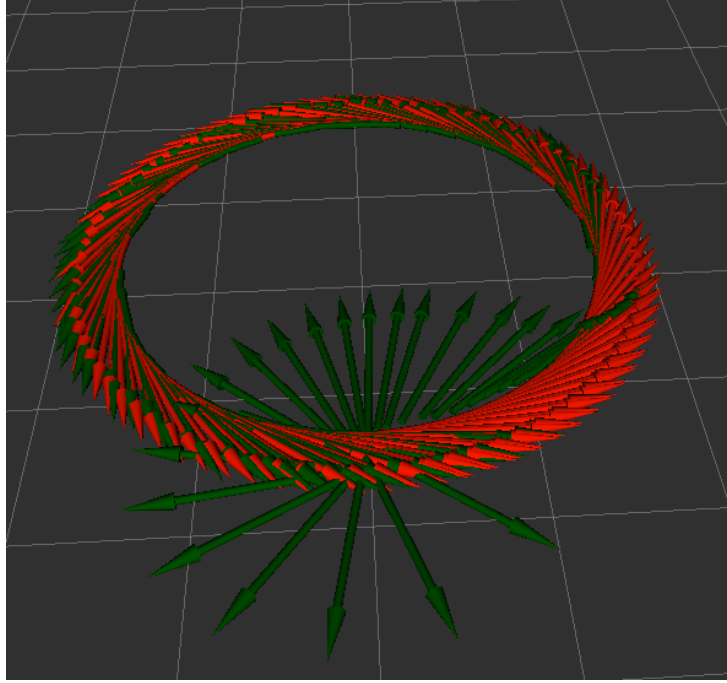


Figure 2: Case B: Larger initial uncertainty with similar steady-state performance

As predicted, the effect of increased initial covariance is limited to the transient phase.

6.3 Case C: Reduced Measurement Noise Covariance

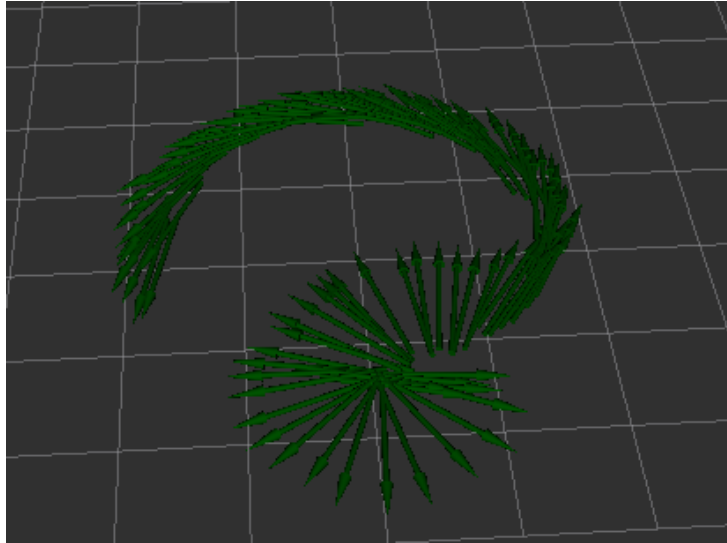


Figure 3: Case C: Overconfident measurement model leading to visible jitter

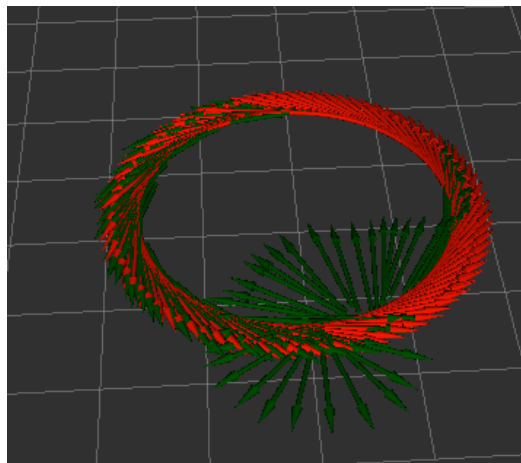


Figure 4: Case C: more trust on noisy measurements

The estimate closely follows noisy measurements, confirming the increased trust in sensor data.

6.4 Case D: Increased Process Noise Covariance

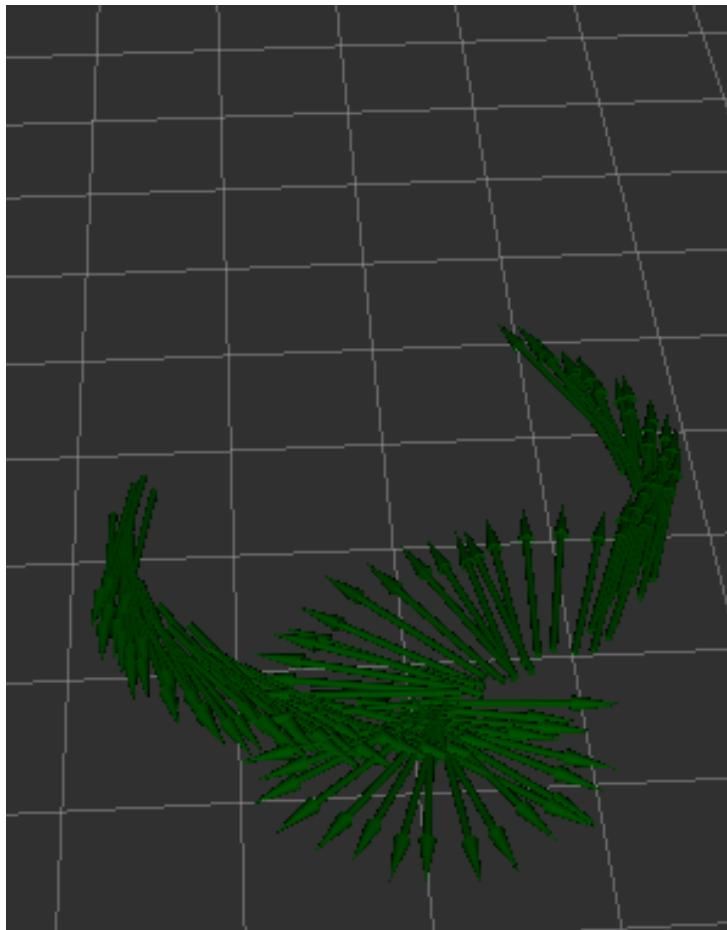


Figure 5: Case D: Conservative motion model producing smoother corrections

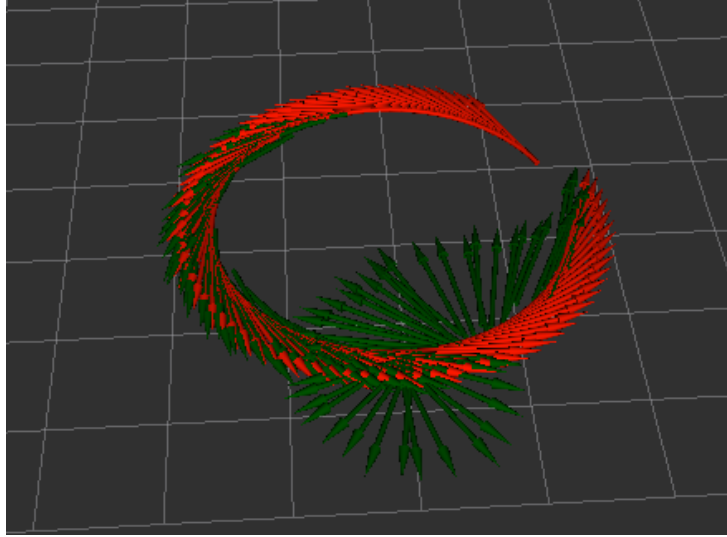


Figure 6: Case D

The trajectory appears smoother than Case C, consistent with reduced reliance on the motion model.

7 Ideality vs Real-World Limitations

While the implemented UKF demonstrates correct estimation behavior in simulation, several simplifying assumptions were made that deviate from real-world robotic systems. These assumptions were intentionally chosen to isolate and study the effects of individual parameters, but they limit direct applicability to real hardware without further modification.

7.1 Constant Process and Measurement Noise

In this implementation, both the process noise covariance Q and the measurement noise covariance R are assumed to be constant over time. In practice, this assumption rarely holds.

Process uncertainty can vary significantly due to changing surface conditions, wheel slip, actuator saturation, or external disturbances. Similarly, sensor noise characteristics may change with temperature, vibration, lighting conditions (for vision-based sensors), or sensor aging. A constant noise

model cannot capture such time-varying effects.

Despite this limitation, constant Q and R matrices provide a reasonable first-order approximation and allow clear interpretation of filter behavior during tuning and analysis.

7.2 Gaussian and Uncorrelated Noise Assumption

All noise sources in this project are modeled as zero-mean Gaussian distributions with no cross-correlation between state variables. This assumption is motivated by the Central Limit Theorem, as real-world uncertainty often arises from the accumulation of many small independent effects.

However, in real systems, noise may be:

- Non-Gaussian (e.g., heavy-tailed or biased)
- Correlated across state variables
- Affected by systematic biases such as encoder drift or sensor misalignment

Ignoring cross-correlation simplifies implementation but may lead to overconfidence or suboptimal estimates in complex environments.

7.3 Perfect Model Structure

The motion model used in the UKF is an ideal differential drive kinematic model. This assumes perfect wheel contact, no slip, and accurate control execution.

In reality, the true system dynamics may include:

- Wheel slip and skidding
- Unequal wheel radii
- Actuator delays and saturation
- Unmodeled dynamics at higher speeds

These effects contribute to process uncertainty, which is only approximately captured through the process noise covariance in our project.

7.4 Estimator Timestep Synchronization

The UKF prediction timestep is assumed to match the simulator update rate exactly. This implies:

- No sensor latency
- No asynchronous measurements
- Deterministic timing

In real robotic systems, sensor data often arrives at irregular intervals, and estimation must handle variable or delayed timestamps. Handling such scenarios typically requires timestamp-based prediction and buffering, which are beyond the scope of this project.

7.5 Absence of Sensor Biases

The simulator generates zero-mean noise without persistent bias. In practice, many sensors exhibit bias that accumulates over time, such as gyroscope bias or wheel encoder drift. These biases can cause long-term estimation errors if not explicitly modeled and estimated as part of the state.

The absence of bias modeling simplifies the estimation problem but reduces realism.

7.6 Implications for Real-World Deployment

Due to these assumptions, careful tuning of Q and R becomes critical when transitioning from simulation to real hardware. Incorrect noise modeling can lead to filter inconsistency, slow convergence, or divergence.

Nevertheless, the simplified framework used in this project provides a clear and controlled environment to understand the role of each UKF parameter and forms a solid foundation for extension to more realistic scenarios.